

Topological Reduction Model (TRM): Extended Framework for Möbius-X Dynamics, Informational Collapse, and Planetary-Scale Transitions

Abstract

The *Topological Reduction Model (TRM)* describes how topological waves emerge inside a Möbius-X manifold, propagate across twisted informational bands, reduce complexity ρ , discharge torsion τ , and generate discontinuous transitions ΔS observable in geological stratigraphy, biospheric evolution, geomagnetic cycles, and cosmic positioning. This extended version formalizes the geometry of Möbius-X, derives the conditions for τ -collapse, explains the mechanism of band-switching, and reconstructs Earth's geological and cosmic trajectory as the result of a non-central topological impact. TRM provides a unified, system-level interpretation of planetary and cosmic phenomena as manifestations of informational topology.

1. Geometry of Möbius-X

1.1 Definition of the manifold

Möbius-X is a **two-band informational manifold** with:

- a single twist point Φ ,
- two informational bands B_1 and B_2 ,
- torsion τ distributed along each band,
- complexity ρ as a scalar field on the surface,
- gradient ΔS describing informational stress.

Formally:

$$MX=(B_1,B_2,\Phi,\tau(x),\rho(x),\Delta S(x))$$

1.2 Why Möbius-X is necessary

A standard manifold cannot:

- reverse orientation,
- switch informational direction,
- collapse torsion into a wave,
- produce dual directionality (south $\rightarrow$ north).

Möbius-X can.

2. Formation of a Topological Wave

2.1 Critical conditions

A topological wave forms when:

$$\tau \rightarrow \tau_{\text{crit}}, \rho \rightarrow \rho_{\text{max}}, \Delta S \rightarrow 0$$

The band cannot maintain stability. The torsion τ collapses.

2.2 Nature of the wave

The wave is:

- not energetic,
- not electromagnetic,
- not gravitational.

It is **purely topological**:

$$W = \partial_t \tau$$

A time-derivative of torsion.

2.3 Propagation

The wave:

- travels along B_1 ,
- reaches Φ ,
- switches to B_2 ,
- reverses orientation.

This explains why geological direction $\neq$ cosmic direction.

3. Earth's Lateral Impact

3.1 Earth's position during τ -collapse

Earth was located at:

$$x \in \partial B_1$$

the **edge** of the band.

Not the axis.

3.2 Consequence

The wave:

- did not pull Earth into B_2 ,
- but **deflected** it.

Earth's trajectory shifted:

- **northward in galactic coordinates,**
- **northward in supergalactic coordinates.**

This is measurable today.

4. Geological Signatures of p -Reduction

4.1 Coal as land-based p -collapse

Coal originates from:

p plants $\gg$ p animals

High- p terrestrial ecosystems collapsed first.

Distribution:

- thickest deposits in paleo-Gondwana,
- strongest signal from the **south**.

4.2 Petroleum as marine p -collapse

Petroleum originates from:

p plankton $\gg$ p dinosaurs

Dinosaurs are irrelevant as a source.

Marine ecosystems collapsed differently:

- slower,
- more distributed,
- producing petroleum instead of coal.

5. Biological Selectivity: Dinosaurs vs Mammals

5.1 High- p metabolism of dinosaurs

Dinosaurs had:

- fungal-like metabolic cycles,
- extremely high throughput,
- high informational complexity p .

High p $\rightarrow$ high susceptibility to collapse.

5.2 Mammals

Mammals had:

- low ρ ,
- low metabolic throughput,
- high resilience.

Thus:

WTRM $\Rightarrow$ ecosystem collapse $\Rightarrow$ dinosaur extinction

The wave did not kill dinosaurs directly. It removed their ecological support.

6. Cosmic Displacement

6.1 Galactic displacement

Earth is above the galactic plane:

$z_{gal} > 0$

6.2 Supergalactic displacement

Earth is above the supergalactic plane:

$z_{SG} > 0$

6.3 TRM interpretation

Incoming wave (geology) $\rightarrow$ **south** Outgoing wave (cosmos) $\rightarrow$ **north**

Because:

$W \rightarrow \Phi \rightarrow B^2$

Band switching reverses direction.

7. Geomagnetic Reversals as τ -Relaxation

7.1 Observed reversals

- 3.58 Ma
- 2.58 Ma
- 0.78 Ma

7.2 TRM interpretation

After τ -collapse:

- ρ decreases,
- reversal cycles shorten,

- core convection destabilizes.

Earth is now in:

$t_4 = \Phi$

the **transition phase**.

8. TRM Temporal Cycle

$t_1 \rightarrow t_2 \rightarrow t_3 \rightarrow \Phi$

- t_1 — Permian–Triassic
- t_2 — Cretaceous–Paleogene
- t_3 — present epoch
- $t_4 = \Phi$ — Möbius transition point

Earth is currently in t_4 .

9. Full TRM Geometry

9.1 Summary of the wave

1. τ reached critical threshold.
2. ρ could not increase.
3. ΔS could not dissipate.
4. τ collapsed $\rightarrow$ wave formed.
5. Wave crossed Φ .
6. Wave switched bands.
7. Earth was struck laterally.
8. Earth displaced northward.
9. Geological and cosmic directions differ.
10. Coal, petroleum, extinctions, reversals = ρ -collapse signatures.

10. Core Insight

TRM describes the Universe as:

Topology $\rightarrow$ Information $\rightarrow$ Dynamics

And Earth as:

A planet shaped by a non-central topological wave

This explains:

- geology,
- biosphere,
- geomagnetism,
- cosmic position,
- fossil fuels,
- extinction patterns,
- present instability.

11. Optional Extensions

Możesz rozwinąć:

- mathematical formalism τ , ρ , ΔS
- diagram Möbius-X band switching
- geological reconstruction
- cosmic displacement model